AI-POWERED IT HELPDESK AND INTELLIGENT SUPPORT MANAGEMENT SYSTEM

Internship Project Report

Submitted by	Arnav Kumar
Institute	The LNM Institute of Information Technology
Organization	GlobalLogic
Internship Mentor	Kundal Singh
Date	July 2026

Acknowledgement

I would like to sincerely thank GlobalLogic to provide me with the opportunity and to give me this internship where I worked on a practical software development project in a professional environment.

I am very grateful to my internship mentor, Kundal Singh, for his guidance and helpful feedback. Many important improvements like department based ticket routing, role specific dashboards, secure email based account onboarding, and controlled software authorization for the AI assistant were done as per suggestions by him.

I also thank my college The LNM Institute of Information Technology for providing me the academic foundation that supported my learning during this internship.

Abstract

This report presents the design and implementation of an AI Powered IT Helpdesk and Intelligent Support Management System developed during an internship at GlobalLogic. The application consists of IT support activities such as ticket creation, assignment and resolution, self service knowledge access, user administration, and initial level issue solving using an AI chatbot.

The project uses React and Vite for the frontend, Python with FastAPI for the backend, Microsoft SQL Server for structured relational data, MongoDB for document oriented data, and Google Gemini for generating conversation responses. Security is maintained using through JWT authentication, role based access control, department based ticket visibility, password hashing, and email based password recovery.

An important feature is the Approved Software Registry. The AI chatbot provides a software download link to the user if the software is present and approved in the list of softwares approved by the company. Authorization and URL selection are implemented in backend logic rather than by the language model.

Contents

1. Introduction, Problem Statement and Objectives

The employees of the company keep facing different problems like hardware issues, software issues, network issues, account-access issues, and requests for softwares. Complaining about these issues or problems in person or through informal ways becomes difficult to track, prioritize and then work on it.

This project solves this problem through a centralized helpdesk app which has structured ticket management system and also an AI powered chatbot. The AI reduces repetitive initial level interaction and sending basic issues and questions to the IT team, while the issues that it is not able to solve is forwarded to the human agent.

1.1 Objectives

- Build a full-stack IT helpdesk using React and Python/FastAPI.
- Implement JWT authentication and role-based access control.
- Support ticket creation, filtering, assignment, status management and resolution.
- Route tickets by support department and restrict agent visibility accordingly.
- Provide role-specific dashboards, also an End User dashboard limited to the user's own tickets.
- Integrate Google Gemini for conversational first-level IT suport.
- Use a Knowledge Base to ground support responses.
- Restrict software download assistance to a company-approved registry.
- Provide email-based account onboarding and password recovery.
- Allow authorized ticket data to be exported to Excel.
- Run the complete system consistently through Docker Compose.

2. Project Scope

2.1 In Scope

- Four roles: Administrator, IT Manager, IT Agent and End User.
- Department-based ticket routing and scoped visibility.
- Ticket lifecycle management and dashboards.
- AI assistant with persistent conversation history.
- Knowledge Base search and AI integration.
- Approved Software Registry with administrative CRUD.
- Email-based temporary credentials and password reset.
- Excel ticket export.
- SQL Server and MongoDB health monitoring.
- Dockerized local deployment.

2.2 Out of Scope

- Formal enterprise ITIL/SLA automation.
- Custom LLM training or fine-tuning.
- Production penetration testing or security certification.
- Native mobile applications.
- Auto procurement/licensing-system synchronisation.

3. Technology Stack

Layer	Technology	Purpose
Frontend	React 18, Vite, React Router, Axios, CSS Modules	SPA, routing and API integration
Backend	Python 3.11, FastAPI, Uvicorn, Pydantic	REST APIs, validation and business logic
Relational Data	Microsoft SQL Server, SQLAlchemy, Alembic	Users, roles, tickets, departments, software and reset tokens
Document Data	MongoDB, PyMongo	AI conversations and Knowledge Base
AI	Google Gemini / google-genai	Conversational first-level support
Security	JWT, python-jose, passlib/bcrypt	Authentication and password security
Reporting	openpyxl	XLSX ticket export
Infrastructure	Docker Compose, Nginx	Container orchestration and frontend serving

4. System Architecture

This application follows a layered architecture. The React frontend will communicate with FastAPI through REST APIs. Route handlers will process HTTP input and authentication dependencies, services contain business rules. The repositories will isolate SQL Server and MongoDB access.

```
Browser / React
        |
        v
FastAPI REST API
   | Routes
   | Services
   | Repositories
   |          |
   v          v
SQL Server   MongoDB
      \       /
       AI Service  ---> Google Gemini
```

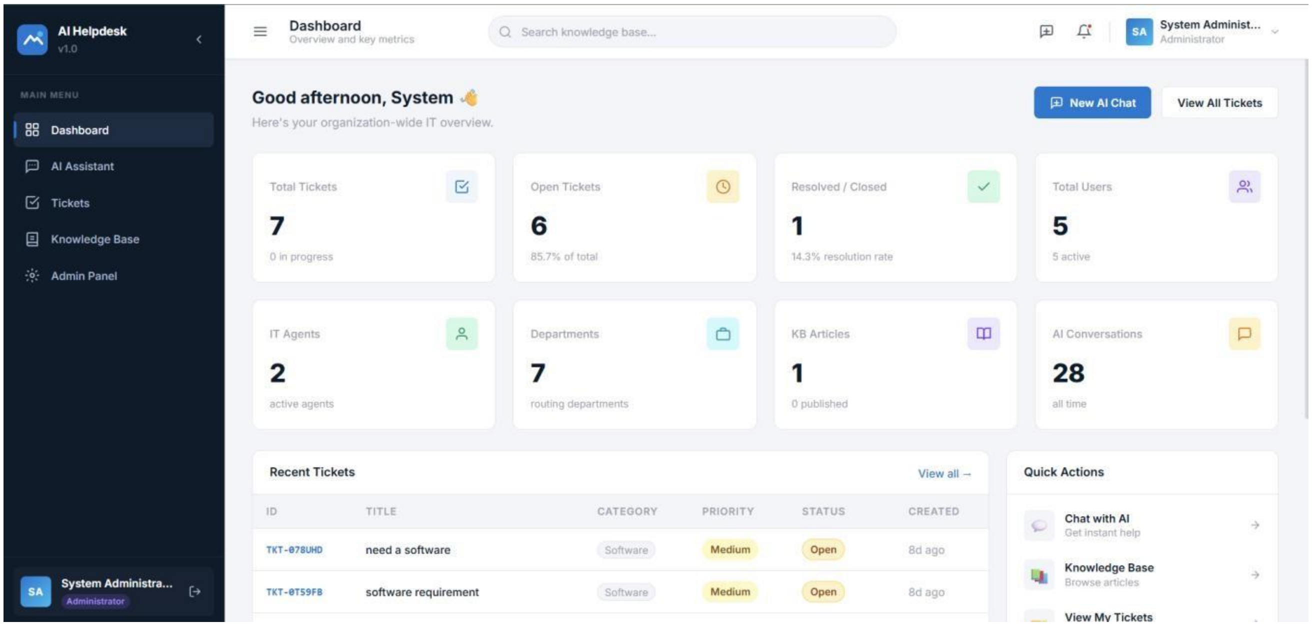


Figure 1: Running AI Helpdesk application

5. Database Design

There are two databases which are being used in this project. This is because the application contains both highly structured relational data as well as flexible document data.

Entity	Database	Reason
Users, Roles	SQL Server	Structured identity relationships
Tickets, Departments	SQL Server	Foreign keys, filtering and assignment
Approved Software	SQL Server	Structured authorization registry and trusted URLs
Password Reset Tokens	SQL Server	Expiry and user linkage
AI Conversations	MongoDB	Variable-length message arrays
Knowledge Base	MongoDB	Flexible text, tags and content

6. User Roles, RBAC and Ticket Management

Role	Primary Responsibility
Administrator	System administration, users and software registry
IT Manager	Broad operational ticket oversight
IT Agent	Department-scoped ticket handling
End User	Raise and view own tickets; use self-service support

Frontend controls adapt to the user's role, but final authority is implemented in the backend. IT agents can only access the tickets which are assigned to their department, while end users can only see the tickets that they created.

6.1 Ticket Workflow

- Create tickets with title, description, category, priority and department.
- Search and filter tickets.
- Assign tickets where the user's role permits it.
- Track Open, In Progress, Pending User, Resolved and Closed states.
- Preserve creation, update and resolution timestamps.

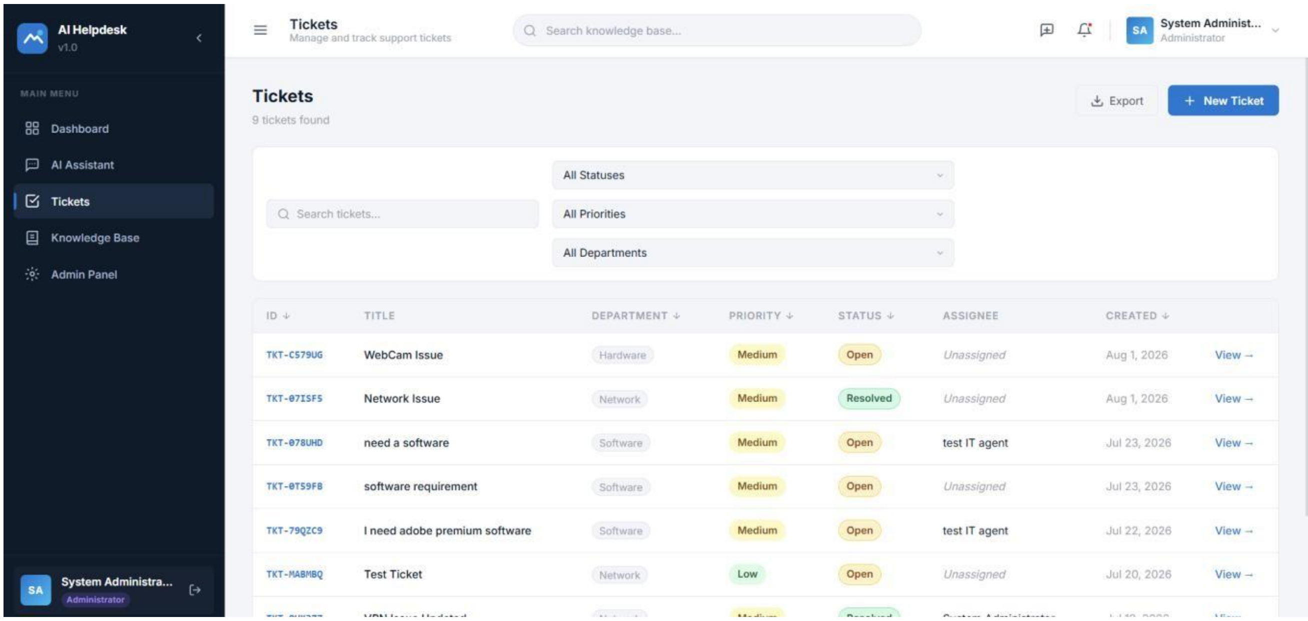


Figure 2: Ticket management interface

7. Role-Specific Dashboards

The dashboard was designed in a way such that each role receives information according to its responsibilities. An important change made the End User dashboard only show user's tickets rather than entire organization's statistics.

Role	Dashboard Focus
Administrator	Administration and system/database health
IT Manager	Broad ticket workload
IT Agent	Department-relevant workload
End User	Own ticket statistics and activity

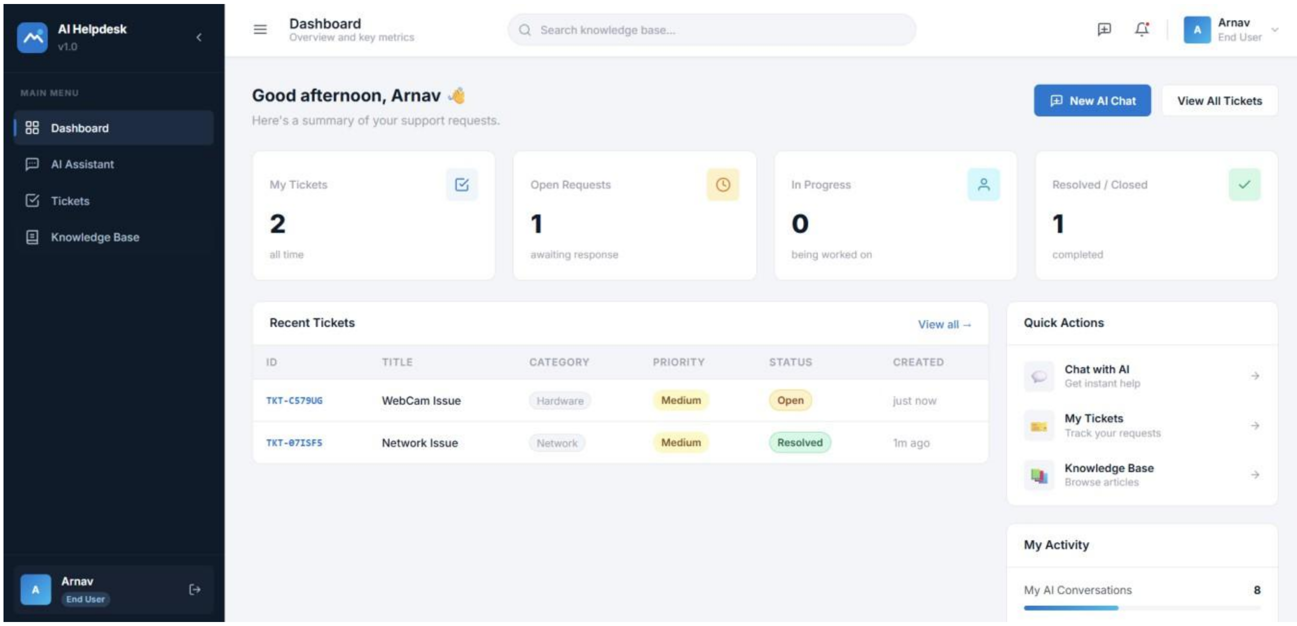


Figure 3: End User dashboard

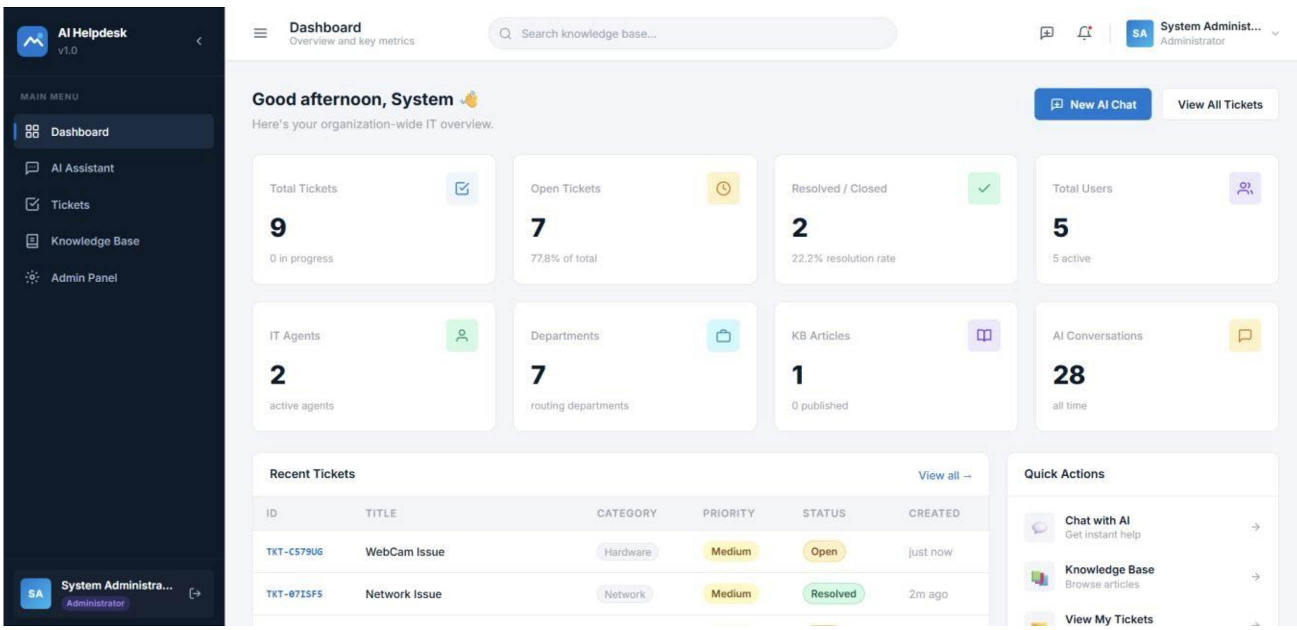


Figure 4: Administrator dashboard

8. AI Assistant and Knowledge Base

Google Gemini helps in first level problem solving through the chatbot. Conversation history is stored in MongoDB so follow up messages can have previous context. Relevant Knowledge Base content can be given to the assistant to answer according to organization's support information.

1. User submits a support message.
2. The backend resolves and stores the conversation.
3. Relevant Knowledge Base context is retrieved where available.
4. Security sensitive software requests are checked through deterministic backend rules.
5. Basic support questions are answered through Gemini.
6. The final response is persisted and returned to the frontend.

AI handles language and assistance, while authentication, authorization, persistence and organizational policy remain in backend code.

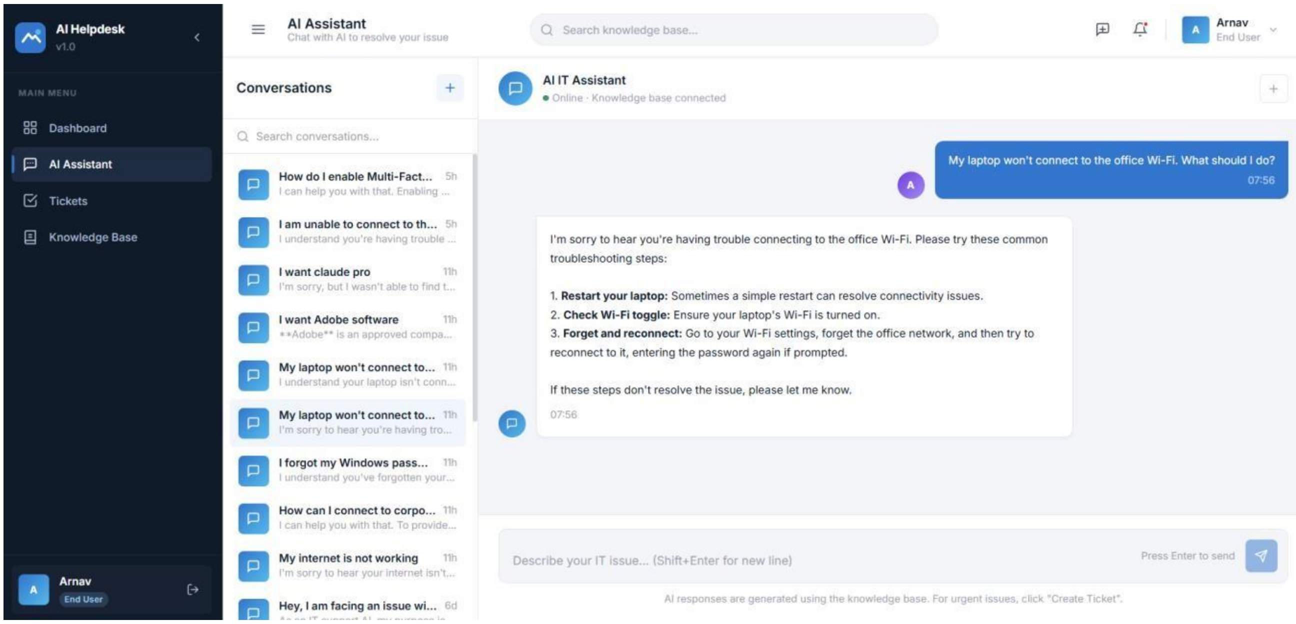


Figure 5: AI-powered first-level support

9. Approved Software Registry and Secure AI Authorization

The Approved Software Registry ensures that the AI assistant provides software download links to the user only if the software has been approved by the company. Administrators maintain software names, aliases, approval state and trusted URLs.

9.1 Authorization Flow

1. A software request is identified.
2. The requested software name or alias is looked up in SQL Server.
3. If an approved record exists, the backend returns the URL stored in that record.
4. If the software is absent or unapproved, the system denies the request without providing a link.
5. A registry lookup failure fails closed and directs the user to IT.

- Gemini does not have the authority for software approval.
- Only the registered URLs are used for approved downloads.
- Aliases use informal names such as 'chrome'.
- Unknown or unapproved software are denied.
- Download URLs are limited to normal HTTP/HTTPS schemes.

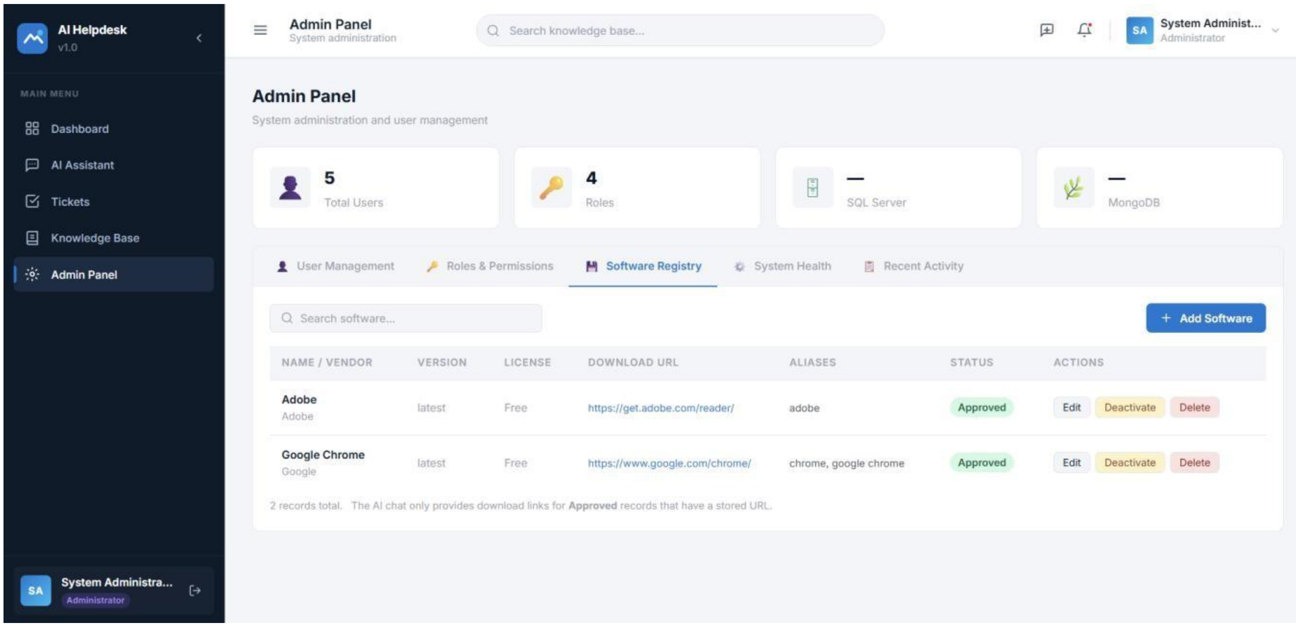


Figure c: Approved Software Registry

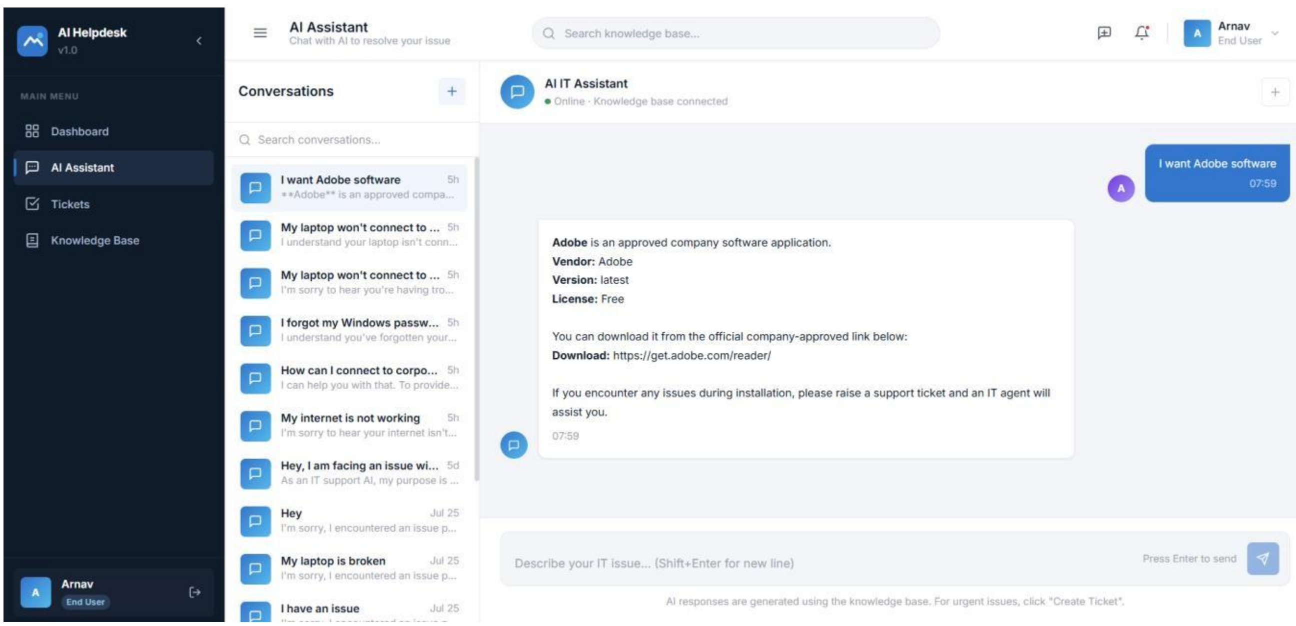


Figure 7: Approved software request

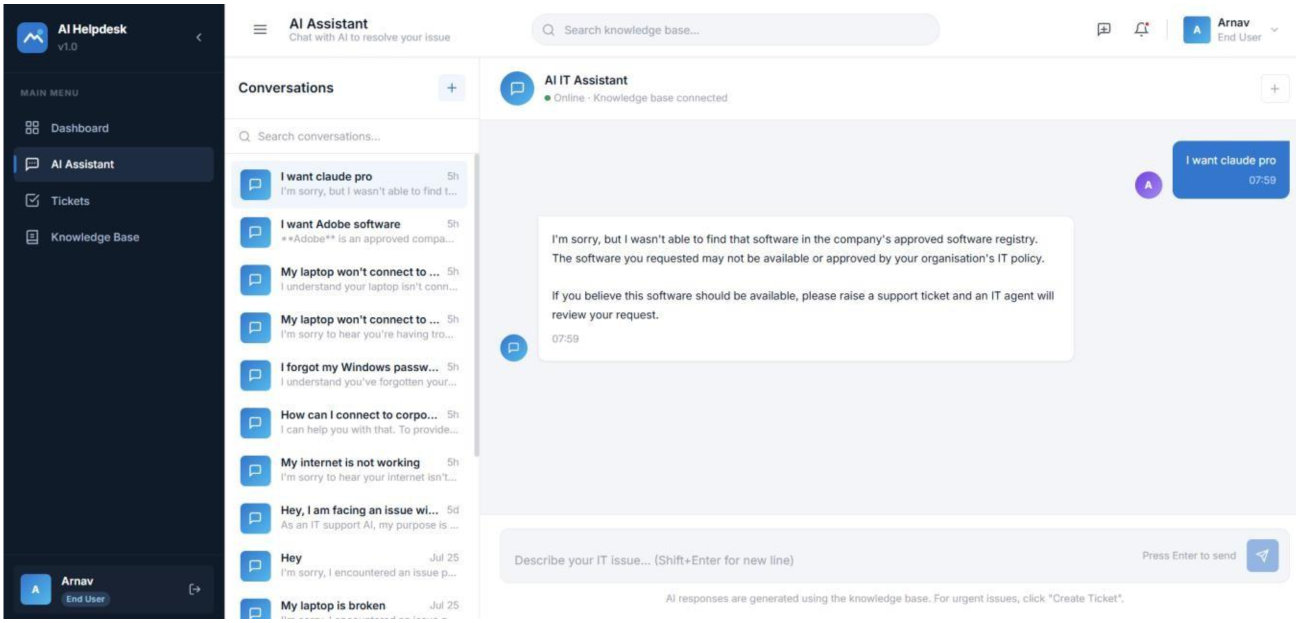


Figure 8: Unapproved software request

5. Authentication, Email Onboarding and Password Recovery

Users authenticate themselves with email and password. Passwords are hashed with bcrypt and not stored as plaintext. Successful login gives a JWT access and refresh tokens, whereas the protected routes then validate the current token and role.

a. New User Onboarding

When an Administrator creates a user, the backend generates a secure password, stores only its hash, and emails the password to the user on their mail address through SMTP.

b. Forgot / Reset Password

1. The user enters an email on the Forgot Password page.
2. The backend returns a generic response to reduce email-enumeration risk.
3. For a valid account, a time-limited reset token and email link are generated.
4. The user submits a new password through the reset page.
5. The backend validates expiry and single-use state before updating the password hash.

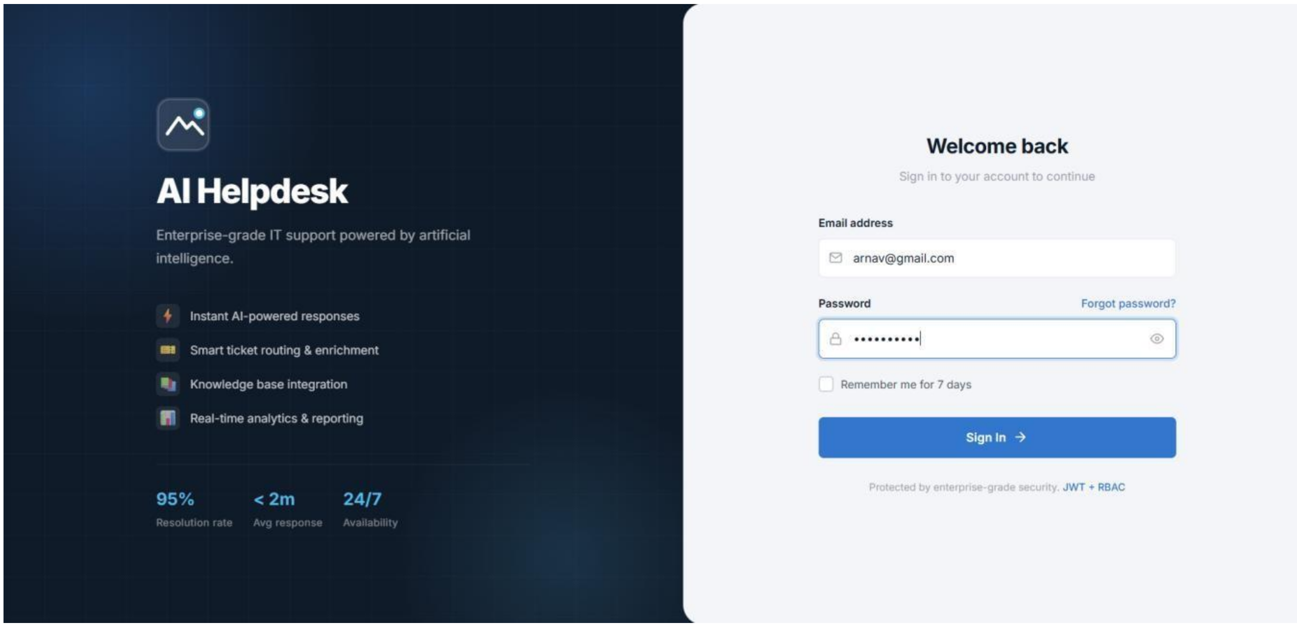


Figure 5: Authentication interface

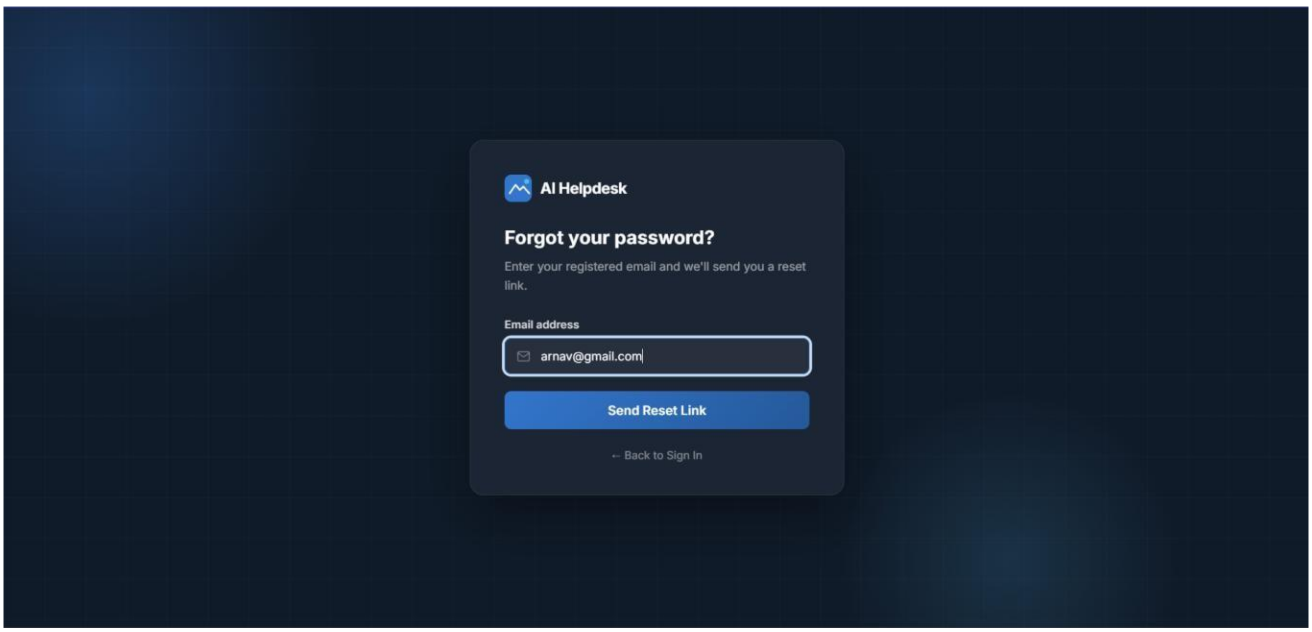


Figure 10: Email-based password recovery

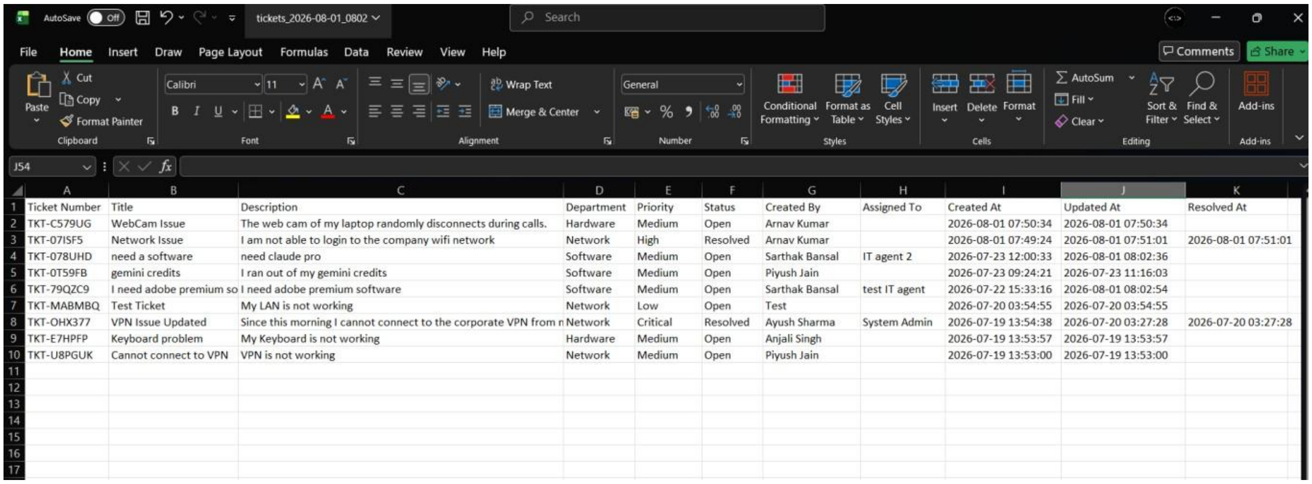
6. Excel Export and Administration

The users who are authorised can export ticket information to excel using openpyxl. The export reuses role scoped ticket retrieval, preventing reporting from bypassing normal visibility boundaries. Active filters are forwarded so the file reflects the selected ticket view.

a. Administration

i. Create and manage user accounts.
ii. Assign roles and departments.
iii. Activate or deactivate accounts.

iv. Maintain the Approved Software Registry.
v. View database/service health information.



Ticket Number	Title	Description	Department	Priority	Status	Created By	Assigned To	Created At	Updated At	Resolved At
TKT-C579UG	WebCam Issue	The web cam of my laptop randomly disconnects during calls.	Hardware	Medium	Open	Arnav Kumar		2026-08-01 07:50:34	2026-08-01 07:50:34	
TKT-07ISF5	Network Issue	I am not able to login to the company wifi network	Network	High	Resolved	Arnav Kumar		2026-08-01 07:49:24	2026-08-01 07:51:01	2026-08-01 07:51:01
TKT-078UHD	need a software	need claude pro	Software	Medium	Open	Sarthak Bansal	IT agent 2	2026-07-23 12:00:33	2026-08-01 08:02:36	
TKT-0T59FB	gemini credits	I ran out of my gemini credits	Software	Medium	Open	Piyush Jain		2026-07-23 09:24:21	2026-07-23 11:16:03	
TKT-79QZC9	I need adobe premium so	I need adobe premium software	Software	Medium	Open	Sarthak Bansal	test IT agent	2026-07-22 15:33:16	2026-08-01 08:02:54	
TKT-MABMBQ	Test Ticket	My LAN is not working	Network	Low	Open	Test		2026-07-20 03:54:55	2026-07-20 03:54:55	
TKT-OHX377	VPN Issue Updated	Since this morning I cannot connect to the corporate VPN from n	Network	Critical	Resolved	Ayush Sharma	System Admin	2026-07-19 13:54:38	2026-07-20 03:27:28	2026-07-20 03:27:28
TKT-E7HPFP	Keyboard problem	My Keyboard is not working	Hardware	Medium	Open	Anjali Singh		2026-07-19 13:53:57	2026-07-19 13:53:57	
TKT-U8PGUK	Cannot connect to VPN	VPN is not working	Network	Medium	Open	Piyush Jain		2026-07-19 13:53:00	2026-07-19 13:53:00	

Figure 11: Excel ticket export

7. Dockerized Runtime

Docker Compose runs the frontend, backend, SQL Server and MongoDB together on a private network.

Service	Port	Purpose
Frontend / Nginx	3000	Serves React production build
FastAPI Backend	8000	REST API
SQL Server	1433	Relational database
MongoDB	27017	Document database

For normal demonstration use, starting the Docker Compose application is enough. A separate Vite development server is required to be started only when actively developing the frontend and may use port 3001 when Docker already occupies port 3000.

8. Development Methodology and AI-Assisted Engineering

Development was done step by step. Generative AI tools including ChatGPT, Claude and Gemini were used for finalising the architecture, assistance on implementation, debugging, review and documenting. The changes were then tested in the running application rather than assuming them to be correct.

1. Foundation and dual-database connectivity.
2. Authentication, users, roles and RBAC.
3. Ticket lifecycle and frontend integration.
4. Gemini assistant and Knowledge Base.
5. Department routing and role-specific dashboards.
6. Excel export and UI corrections.
7. Email onboarding and password recovery.
8. Approved Software Registry and secure software authorization.
9. Debugging and stabilization.

10. Limitations and Future Enhancements

a. Limitations

- Internship-scale implementation, not tested for enterprise traffic.
- SMTP features depend on an external mail provider.
- Software Registry requires manual maintenance.
- AI availability depends on the external Gemini API.
- Knowledge Base quality depends on authored content.
- Large Excel exports are generated synchronously.

b. Future Enhancements

- SLA tracking and automatic escalation.
- Ticket analytics and workload visualization.
- Email notifications for ticket status changes.
- File attachments and richer audit trails.
- Microsoft Entra ID / enterprise SSO.
- Semantic/vector Knowledge Base retrieval.
- Asynchronous export for large datasets.
- Azure deployment and monitoring.
- Automated unit and integration testing.

11. Conclusion

This project shows how a regular full stack IT support system can be improved by integrating generative AI without making the AI responsible for security related decisions. React and FastAPI provide the application layers, SQL Server and MongoDB provide with persistence needs, and Docker Compose provides a runtime.

One of the most important architectural thing is that conversational intelligence is separated from organinsational policies. Gemini improves initial level support interaction, whereas identity, RBAC, department isolation, ticket access and approved software logic are implemented in the backend logic.

This internship also gave me a practical experience in integrating different technologies, responding to mentor feedback, debugging issues encountered during runtime, and using AI development tools.